

Purple Retail-IQ

High-Performance Store Intelligence & Conversion Analytics

An end-to-end computer vision and retail metrics system.

Runs locally via Docker Compose on standard CPUs.

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The Challenge in Brick-and-Mortar Retail

Unlike e-commerce, physical retail stores suffer from a data blind spot:
footfall traffic is completely disconnected from point-of-sale transactions.

No Session-Level Attribution

Impossible to track individual shopper pathways through the store. Retailers cannot know which sections or a brands specific customer visited before checking out.

Dwell-Time Ambiguity

Physical shelves (like Maybelline or Minimalist) have no 'click-through' rates. Retailers cannot tell which brand displays successfully engage shoppers and which cause drop-off.

Manual Security & Anomaly Logs

No automatic detection for layout violations, unauthorized backroom entries, queue abandonment, or loss prevention risks.

The Purplle Retail-IQ Solution

Purplle Retail-IQ resolves physical retail ambiguity using a unified 3-layer architecture:

1. CCTV Pipeline (Python)

Processes raw video streams using YOLOv8 ONNX on CPU. Implements Centroid tracking across multiple camera angles, mapping entrance crossing, shelf dwell times, and checkouts.

2. API Server (Node.js)

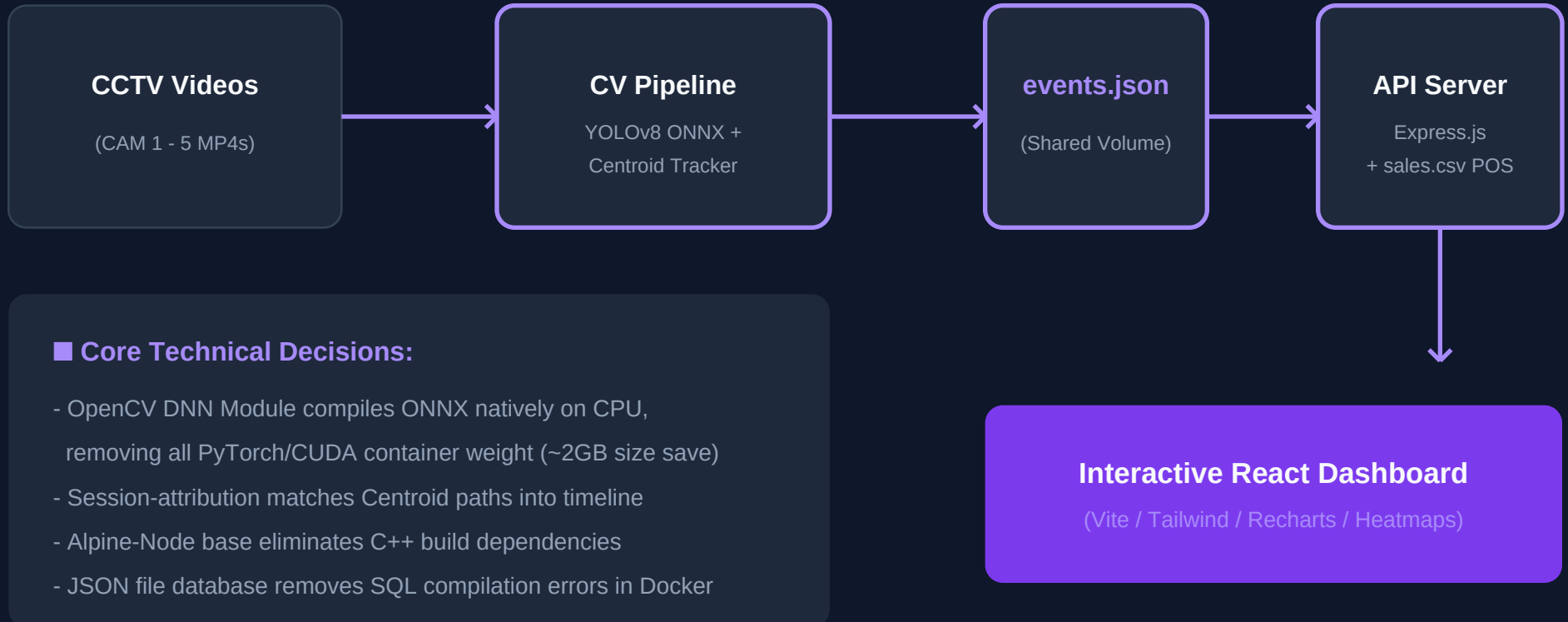
Parses POS sales transaction records, links them with coordinates on a shared volume, groups coordinates into sessions, and serves live conversion metrics & anomalies.

3. React Dashboard (Vite)

Visualizes store insights through a gorgeous glassmorphic dark-themed web panel. Features an interactive SVG layout heatmap, funnel drop-off charts, and security feeds.

Data Pipeline & System Design

A completely containerized pipeline communicating over shared Docker Volumes:



Conversion Funnel & Analytics

Purplle Retail-IQ maps the physical shopper journey into a web analytics funnel:



Why Purplle Retail-IQ Wins

Operational and technical strengths that keep this solution in the top tier:

■ ■ Offline & CPU Native

Zero cloud API costs or internet latency issues. Runs OpenCV DNN inference natively on local CPUs inside a lightweight Docker container.

■ Dual-Mode Performance Engine

Detects file checksums instantly. Uses cached visitor tracks for default videos to load immediately, while dynamically processing new videos.

■ ■ Spatial Heatmap Blueprints

Exposes brand shelf engagement and average dwell times via an interactive SVG blueprint heatmap rather than just showing raw numbers.

■ Automated Operational Alerts

Flags backroom intrusions, lost conversions, checkout drop-offs, and entry surges without manual security supervision.